MIL-H-36203A
17 September 1973
SUPERSEDING
MIL-H-36203(DM)
21 May 1964

MILITARY SPECIFICATION

HANGER, X-RAY FILM, Processing, Dental

1. SCOPE

1.1 This specification covers a clipless hanger having a capacity of 16 films, suitable for use in processing 1-1/4 by 1-5/8 inch dental x-ray films.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-566 | Box, Folding, Paperboard |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |
| PPP-B-676 | Boxes, Set-Up, Paperboard. |

Military

| | |
|---|---|
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible. |

FSC 6525

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STANDARDS

Federal

Fed. Std. No. 66 Steel; Chemical Composition and Hardenability.
Fed. Test Method Std. No. 151 Metals, Test Methods.

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Material. All metal parts of the hangers, including film holding sections, hanger frame, hooks, handle, rivets and latch shall be made entirely of AISI 301, 302 or 304 corrosion-resisting steel, in accordance with FED-STD-66. All metal parts shall pass the chemical resistance tests specified in 4.4.2. A corrosion spot visible to the unaided eye, any pitting, or etching on any significant surface on any of the hangers subjected to these tests shall be cause for rejection, with the exception of slight corrosion at the spot welds and pivots. Hardness shall be a minimum of B75 on the Rockwell scale, except for rivets.

3.2 Design and construction. The clipless film hanger shall be constructed in accordance with Figure 1. The hanger shall be designed to accommodate 16 standard 1-1/4 by 1-5/8 inch dental X-ray films and shall consist of two film holding sections pivotally mounted in a spot welded frame and handle assembly.

3.3 Operation. With the hanger held horizontally in the closed position (and the handle oriented, as shown on figure 1) opening of the hanger by releasing the latch, shall cause each hanger section to rotate approximately 90° and come to rest in an upright position convenient for film loading. With the hanger in the open position, it shall be possible to rest the bottom members of the two film holding sections on a flat surface to provide access and support for loading and unloading individual films into and out of the hanger. In this position (open), the major axis of the oval slot in the film holding members shall be parallel to each other. The size and contour of the slots shall be such that standard 1-1/4 by 1-5/8 inch dental X-ray films can be easily inserted through the vertically aligned slots in the holders and come to rest against the bottom members.



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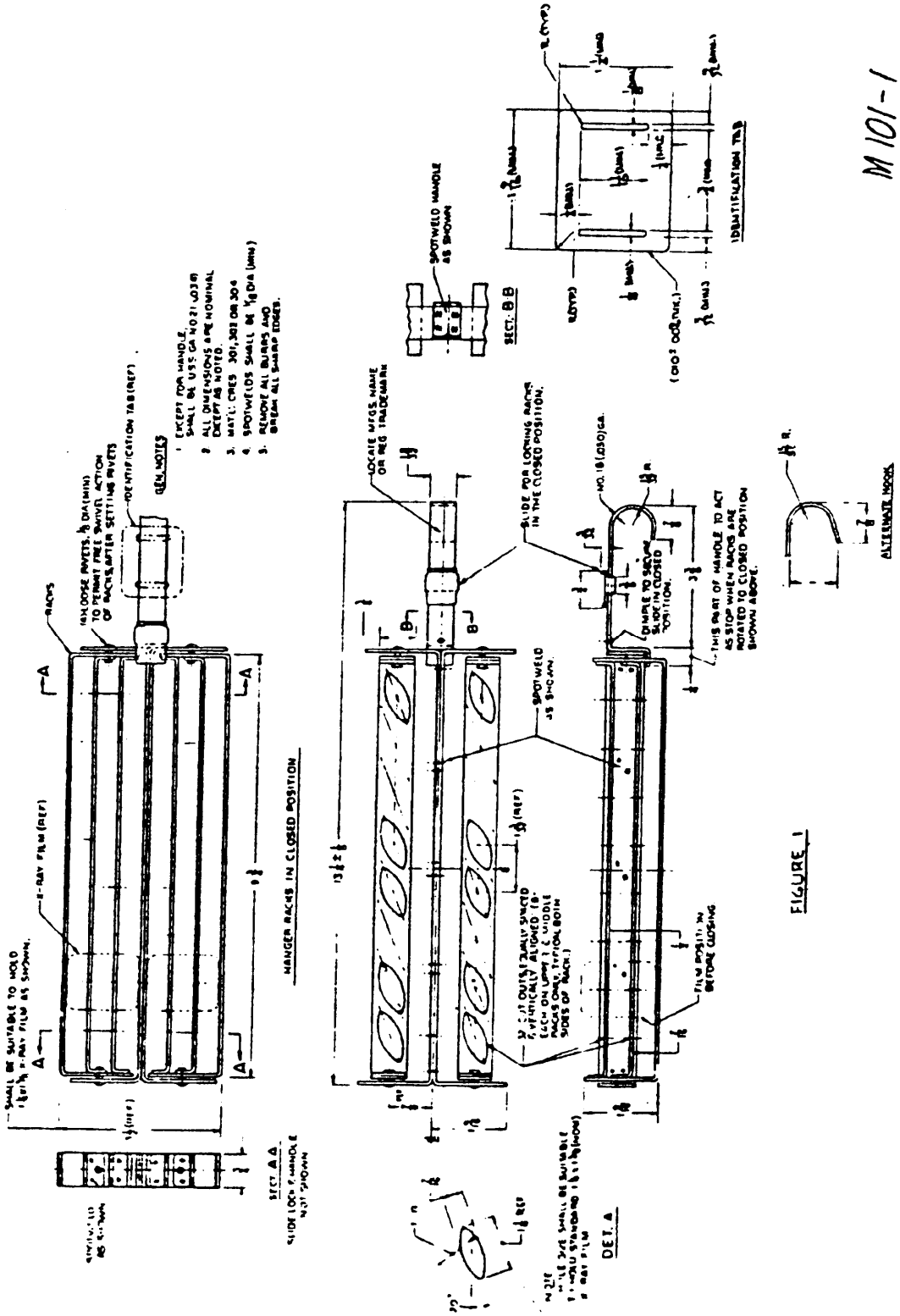



FIGURE 1



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3.3 Operation, cont'd.

The films shall be retained within the aligned oval slots without bowing. When the hanger is closed by rotating the film holding sections approximately 90° against the stop shown in the drawing, the bottom (unperforated) members of the film holding sections shall face outward and the free ends of the inserted films prevented from coming out of the hanger. A latch as shown on the drawing, shall be provided to retain the film holding sections in the closed position. The angulation of the films, in the holding section of one side of the closed hanger shall be substantially equal to but in opposition with the corresponding angulation of the films in the other holding section. The hanger shall meet the requirements of the functioning test specified in 4.4.3.

3.4 Identification tabs. Each hanger shall be furnished with three white cellulose acetate or polyvinyl tabs. Each tab shall have two cut-outs to permit mounting on the handle. Each tab shall have a matte finish on at least one of its surfaces to permit pencil marking. One tab shall be mounted on the handle and the other two shall be mounted in two of the vertically aligned oval slots in the film holders. Each tab shall be dimensioned as shown on Figure 1.

3.5 Identification marking. The hangers shall be legibly and permanently marked with the name or registered trademark of the manufacturer, and year of manufacture.

3.6 Passivation. The hanger shall be treated to remove contamination and to restore the passivity of the base material. As an alternate, the hangers may be electropolished.

3.7 Workmanship. Workmanship shall be first class throughout. Hangers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Govenment's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Classification of Inspections.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table I.

Table I

| | Inspection level | AQL (defects per hundred units) |
|---|---|---|
| For visual examination | | |
| Major defects | II | 1.5 |
| Minor defects | II | 4.0 |
| For dimensional examination | S-1 | 2.5 |

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Standard MIL-STD-105 and Table II.



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Table II

| Component (unit of product) | Characteristic | Requirement | Test procedure | Inspection level | AQL (percent defective) |
|---|---|---|---|---|---|
| Raw material | Chemical Analysis | 3.1 | 4.4.1 | * | * |
| * End item | Chemical resistance to developer | 3.1 | 4.4.2.1 | S-2 | 4.0 |
| | Chemical resistance to fixer | 3.1 | 4.4.2.2 | S-2 | 4.0 |
| | Functioning Test | 3.3 | 4.4.3 | S-2 | 2.5 |
| Exterior shipping container and contents | Packaging<br>Packing<br>Marking<br>Levels of protection | ** | | S-1 | 4.0 |

* Inspection level and AQL not applicable.

** See procurement document and applicable paragraphs of section 5.

4.3 Examination. The hanger shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects as indicated in Table III.

Table III Categories and defects*

| Major | |
|---|---|
| 101 | Oval slots, edges and corners of the hanger components not free of burs and sharpness. |
| 102 | Hanger not complete with all specified component parts. |
| 103 | The holding sections fail to swivel freely (in the open position of the hanger). |
| 104 | Rivets not properly driven or spun. |
| 105 | Spot-welded joints not secure and free of flash, cracks or discoloration. |
| 106 | Mating parts not properly fitted and aligned. |

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| | |
|---|---|
| 107 | Component parts in the finished hanger not free of cracks, and unspecified bends. |
| 108 | Finish not smooth and free of pits or corroded areas. |
| 109 | Lock not secure in closed position. |
| Minor | |
| 201 | Identification marking not present or not permanent and legible. |
| 202 | Hangers not clean and free of oil, chips or dirt. |

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The hangers shall be examined for defects in dimensions. Any dimensions not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical analysis. Chemical analysis shall be conducted in accordance with test methods 111.2 or 112.2 in Federal Test Method Standard No. 151.

4.4.2 Chemical resistance to processing solutions.

4.4.2.1 Chemical resistance to developer. Prepare developing solution in accordance with the following formula:

| | |
|---|---|
| Metol | 2.2 grams |
| Sodium sulfite (desiccated) | 72.0 grams |
| Hydroquinone | 8.0 grams |
| Sodium carbonate (mono-hydrate) | 56.1 grams |
| Potassium bromide | 4.0 grams |
| Water to make | 1.0 liter |



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Clean hangers to be tested thoroughly by a method that will assure removal of all grease and oil. During the tests the temperature of the developer shall be maintained at 110° ± 5°F. Immerse the hangers to approximately one-half their height in the developing solution for 72 hours, remove, and permit the excess solution to drain off (do not rinse or dry). Load the hangers to capacity and suspend hangers and film in the developer to such depth that the liquid level is at least 1/4 inch above the top of the film. Cover the container so as to confine vapors and maintain at a temperature of 110° ± 5°F. for 72 hours. Discard the film, rinse, dry, and examine the submerged portion of the hanger for pitting or etching and evidence of corrosion.

4.4.2.2 Chemical resistance to fixer. Prepare fixing solution in accordance with the following formula:

| | |
|---|---|
| Water | 600 milliliters |
| Sodium thiosulfate | 240.0 grams |
| Sodium sulfite (desiccated) | 15.0 grams |
| Acetic acid, 28 percent | 48.0 milliliters |
| Potassium alum | 15.0 grams |
| Water to make | 1.0 liter |

Duplicate the two stage test procedure described in 4.4.2.1 using the fixing solution maintained at 110° ± 5°F. in place of the developer with dental films exposed to white light.

4.4.2.3 As an alternate to the above specified developer and fixer solutions standard commercial x-ray chemistry shall be used.

4.4.3 Functioning test. Each sample hanger shall be opened, loaded to capacity with standard 1-1/4 by 1-5/8 inch dental X-ray films and then closed. To meet the requirements of this test, the hanger shall open easily, individual dental films shall load readily, be retained flat within the aligned oval slots and the hanger shall close easily and lock. In the closed position of the hanger none of the films in the holders shall be bowed along either their length or width. Loaded hanger shall be submersed and agitated in developer solution for a one minute period, after which the latch shall be examined to insure it remained locked.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).



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5.1.1 Unit of issue. One hanger as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging Quantities

| Unit package | Intermediate package | Exterior container |
|---|---|---|
| 1 unit | 6 units | 36 units |

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636 type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).



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5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This hanger is intended for holding 1-1/4 by 1-5/8 inch dental film during processing.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Need for first article sample should be considered.

6.3 This specification covers the following item appearing in the Federal Supply Catalog.

| National Stock Number | Item Identification |
|---|---|
| 6525-00-074-4542 | HANGER, X-RAY FILM, Processing, Dental |

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6525-1112

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.



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CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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